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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

January 2018 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-JAN18

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	Master Functional Skills (Model Exemplar)
Centre Name:	Master Functional Skills (MFS01)
Lead Tutor:	Abdul · MasterFunctionalSkills.co.uk
Edition:	2026 Clean Empty-QP Rebuild (Zero Collisions Verified)

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2
Data Book (Examination)

Insert

Instructions

- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/answer book.

4368

QAN 500/8702/2

IB/M/Jan18/E5

Data Sheet for Minibus

Hiring a minibus

Some companies charge

a fixed amount per day

plus

an amount per mile.

Example

A company charges

£37 per day

plus

40p per mile.

Jack hires a minibus for 2 days and travels 358 miles.

2 × £37 = £74

358 × 40p = £143.20

Total charge = £74 + £143.20

= £217.20

Jack has to pay £217.20

Distances between places by road

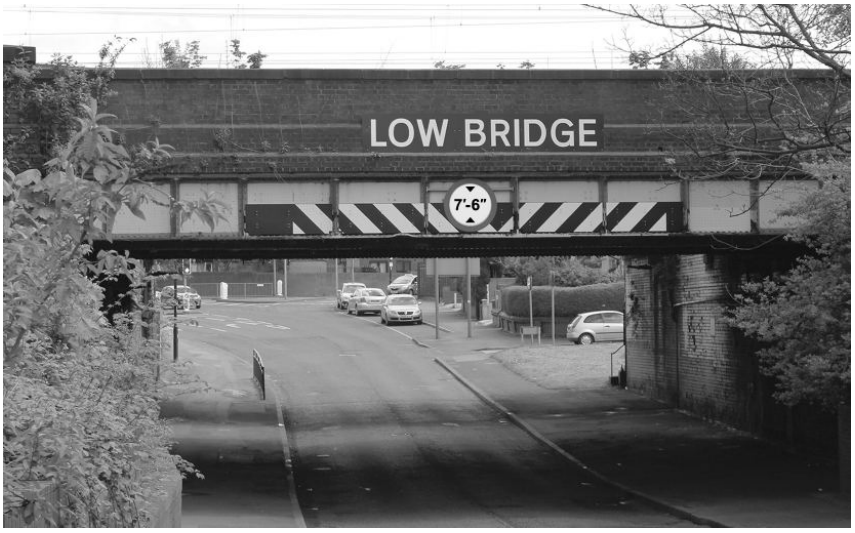
The table shows the distances, in miles, between four places.

	Bristol	**Exeter**	**Southampton**	**Truro**
Bristol		81	100	168
Exeter	81		114	87
Southampton	100	114		198
Truro	168	87	198	

Example

The distance between Exeter and Southampton is 114 miles.

Low bridges



A sign tells drivers the maximum height of a vehicle that can pass under the bridge.



7′ 6″ is 7 feet 6 inches.

1 foot = 12 inches

Converting feet to metres

Use this formula.

$$m = 0.3048f$$

m is the height in metres

f is the height in feet

Data Sheet for Wages

National Minimum Wage

By law, workers should be paid at least the National Minimum Wage.
This table shows the hourly rates for different ages.

Age (years)	16 to 17	18 to 20	21 to 24	25 or over
Hourly rate	£4.05	£5.60	£7.05	£7.50

Piece workers

Some workers are paid for each item they make.
For example, Jenny makes shirts for a clothing company.
She is paid for each shirt she makes.

Fair rate

This is the minimum amount piece workers should be paid for each item they make.

To work out the fair rate

Step 1 Work out mean number of items made per worker in one hour

Step 2 Work out National Minimum Wage ÷ the answer to **Step 1**

Step 3 Increase the answer to **Step 2** by 20%

The answer to **Step 3** is the fair rate per item.

Example

A company employs piece workers to make vests.
All the workers are 18 to 20 years old.
The National Minimum Wage for these workers is £5.60

Step 1 Mean number of vests made per worker in one hour = 16

Step 2 £5.60 ÷ 16 = £0.35

Step 3 Increasing £0.35 by 20% = £0.42

The fair rate is 42p per vest.

END OF DATA

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Answer **all** questions in the spaces provided.

1 Dance Show

Sue runs a dance school.



The students at my dance school are in a show.

Q1(a) Worked Solution:
Mark second grid unless first grid blank
Example of BS
Dance Type
1st Tap Grace
2nd Tap Dita
3rd Ballet Amy
4th Tap Grace
5th Tap Amy
Sue
At least one correct shape drawn on grid

1 (a) The show has four tap dances and one ballet dance.

Each dance has three of these students.

Amy Dita Fiona Grace Leah Mel Tia

- Amy, Dita and Tia are in the 5th dance.
- Amy is **not** in the 1st dance.
- Grace is **not** in the ballet dance.
- Each student is in at least two dances.
- No student is in consecutive dances.

Show one possible plan for the students.

[5 marks]

Question 1(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

At least 6 correct shapes drawn on

At least 8 correct shapes drawn on

[M1] [M1] [M1] [A1] [A2]

Do not write outside the box

Practise on this grid.

Dance	Type	Students		
1st	Tap			
2nd	Tap			
3rd	Ballet			
4th	Tap			
5th	Tap			

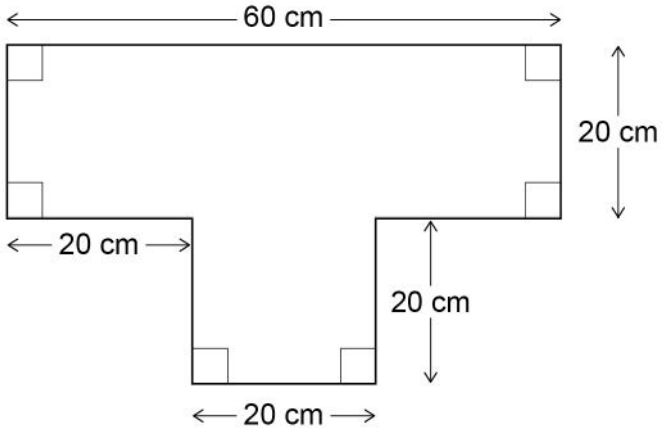
Put your answer on this grid.

Dance	Type	Students		
1st	Tap			
2nd	Tap			
3rd	Ballet			
4th	Tap			
5th	Tap			

Question 1 continues on the next page

1 (b) Sue makes costumes for the show.

She wants to cut out 10 of these shapes from a piece of material.



Not drawn accurately

Each scale drawing opposite shows the piece of material.

Sue says,

"I can cut out 10 of these shapes from this piece of material."

Is she correct?

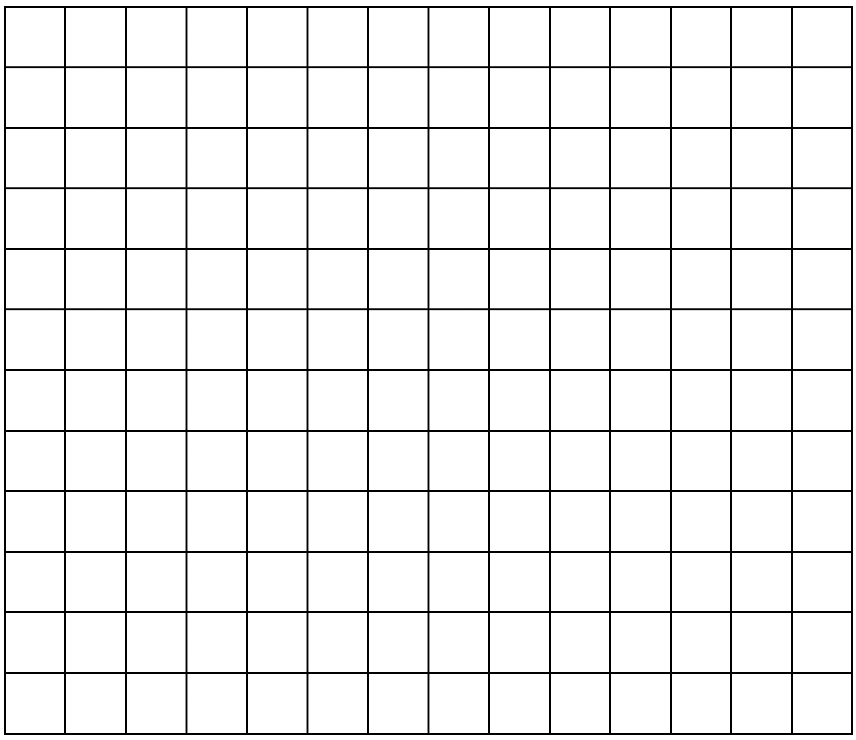
Use the scale drawings to show your working.

[5 marks]

Do not write outside the box

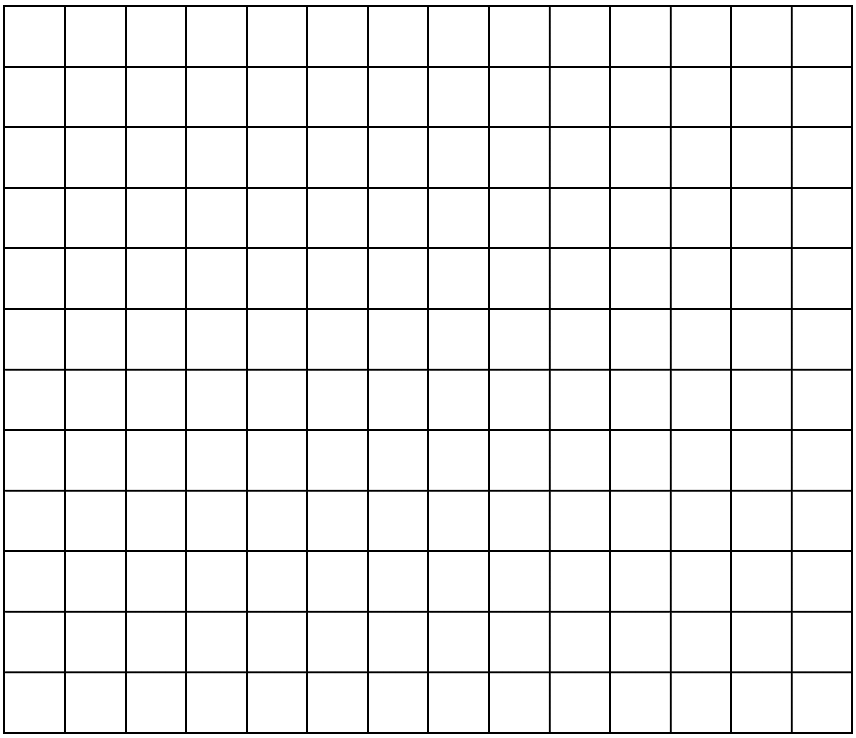
Practise on this scale drawing.

Scale ☐ represents a 10 cm by 10 cm square



Put your answer on this scale drawing.

Scale ☐ represents a 10 cm by 10 cm square



Do not write outside the box

1 (c) The show is on for **two** nights.

For each night, there are

80 adult tickets at £10.50 each

and

60 child tickets at £7.20 each.

The **total** cost of putting on the show for two nights is £925

Sue sells 90% of the adult tickets and $\frac{2}{3}$ of the child tickets.

She says,

''We have made **more than** £1200 profit.''

Is she correct?

You **must** show your working.

[7 marks]

17

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2 **Minibus**

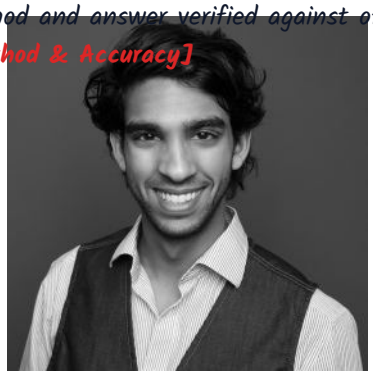
There is a **data sheet** for Minibus.

†2 (a) Samir wants to hire some minibuses to go to a hockey match.

Q2(a) Worked Solution:

Method and answer verified against official AQA mark scheme.

[Method & Accuracy]



I need enough minibuses to carry 96 people.

Samir

Each minibus can carry 15 people.

How many minibuses does Samir need to hire?

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

Question 2 continues on the next page

14 friends are going to Truro for a weekend.

12 of the friends will travel from Southampton.
They will pick up the other 2 friends in Bristol.

2 (b) Circle the total number of miles for the journey to Truro.

Southampton ⟶ Bristol ⟶ Truro

[1 mark]

214 268 298 536

2 (c) The friends look at this advert.

Minibus hire
£28 per day
plus
65p per mile

They drive the minibus for 3 days.

On Friday the minibus will travel

Southampton ⟶ Bristol ⟶ Truro

On Sunday the minibus will travel

Truro ⟶ Bristol ⟶ Southampton

Each friend from Southampton will pay £10 more than each friend from Bristol.

Will each friend from Southampton pay **less than** £35?
You **must** show your working.

[7 marks]

Do not write outside the box

2 (d) On the journey, they come to a low bridge.



The height of the minibus with a loaded roof rack is 3.1 metres.

Will the minibus pass under the bridge?

You **must** show your working.

[4 marks]

15

Do not write outside the box

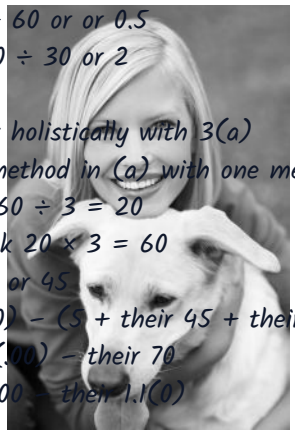
3 **Guide Dogs in training**

Guide Dogs train at a centre each day from Monday to Friday.

In the evenings and at weekends, people look after them at home.

Q3(a) Worked Solution:

30 ÷ 60 or or 0.5
or 60 ÷ 30 or 2
20
Mark holistically with 3(a)
No method in (a) with one method in check
(a) 60 ÷ 3 = 20
Check 20 × 3 = 60
× 60 or 45
9(.00) − (5 + their 45 + their 20)
or 9(.00) − their 70
or 9.00 − their 1.1(0)



Sarah looks after Buddy.

Sarah takes Buddy from her home to the centre.

She then travels to work.

Sarah makes these notes.

or 7.5
[M1] [A1] [M1] [M1]

Home to centre	10 miles average speed 30 miles per hour
At centre	5 minutes
Centre to work	$\frac{3}{4}$ of an hour

Do not write outside the box

†3 (a) How many minutes does Sarah take to travel from her home to the **centre**?

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

3 (b) Sarah needs to arrive at **work** by 9.00 am

Work out the **latest** time that she can leave home.

[3 marks]

Question 3 continues on the next page

Do not write outside the box

The daily amount of food for a dog depends on the weight of the dog.

Weight of dog	Daily amount of food
10 kg	176 g
20 kg	296 g
30 kg	402 g
40 kg	494 g

25 kg is halfway between 20 kg and 30 kg
The daily amount of food for a 25 kg dog is 349 g

Dogs get two feeds each day.

Morning feed	Half the daily amount
Evening feed	Half the daily amount

3 (c) Buddy weighs 35 kg

Sarah uses a cup to measure out the **morning** feed.

1 cup holds 96 g

Sarah says,

"I need to measure out between $2\frac{1}{4}$ and $2\frac{1}{2}$ cups."

Is she correct?

You **must** show your working.

[5 marks]

Do not write outside the box

3 (d) Omar looks after a different dog.

The **daily** amount of food for his dog is 349 g

Omar says,

"A 20 kg bag of food is enough to feed my dog for 60 days.''

Is he correct?

You **must** show your working.

[4 marks]

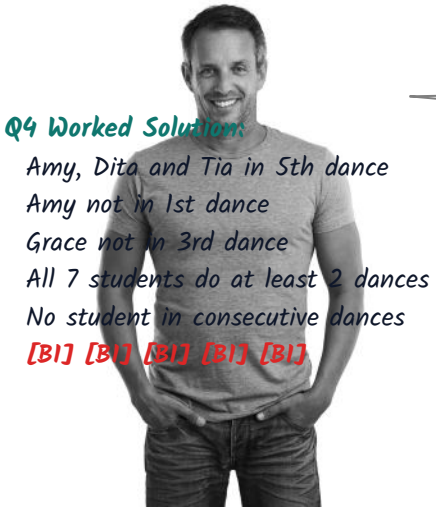
15

Turn over for the next question

Do not write outside the box

4 **Wages**

There is a **data sheet** for Wages.



Bob

The table shows the number of T-shirts 20 piece workers made in one hour.

Number of T-shirts	Number of workers
7	9
8	4
9	5
10	2
Total	20

Do not write outside the box

4 (a) Show that the mean number of T-shirts made per worker is 8

[3 marks]

4 (b) The workers are all 25 years old or over.

Bob says,

''The fair rate is £1.05 per T-shirt.''

Is he correct?

You **must** show your working.

[5 marks]

Do not write outside the box

4 (c) One week, the company needs to make at least 3600 T-shirts.

The mean number of T-shirts each worker makes per hour is 8

Bob will use

10 full-time workers, each working 7 hours per day for 5 days

and

some part-time workers, each working for $4\frac{1}{2}$ hours per day for 4 days.

How many part-time workers does he need?

You **must** show your working.

[5 marks]

13

END OF QUESTIONS

Do not write outside the box

There are no questions printed on this page

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ANSWER IN THE SPACES PROVIDED

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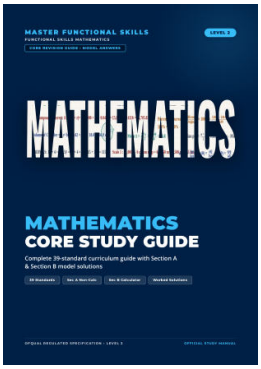
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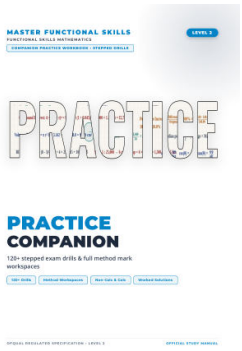


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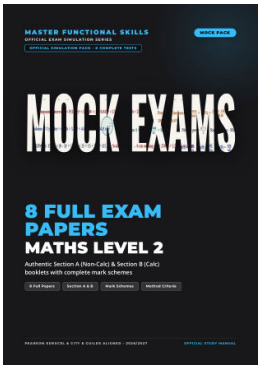


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